Midterm Practice

1. Which of the following single-qubit gates is its own inverse **and** maps $|\mathbf{0}\rangle$ to a superposition state?

- a. X
- b. Z
- c. H
- d. S

H is self-inverse ($H^2 = I$) and maps $|0\rangle \mapsto (|0\rangle + |1\rangle)/\sqrt{2}$, a superposition. X maps $|0\rangle \to |1\rangle$ (not a superposition); S is not self-inverse (S^2=Z); CNOT is two-qubit.

2. The controlled-NOT gate is **not** ...

- a) unitary
- b) reversible
- c) its own inverse
- d) diagonal

CNOT is unitary, reversible, and its own inverse ($CNOT^2=I$), but not diagonal in the computational basis.

3. In the Deutsch–Jozsa algorithm with an n-bit input, the final measurement of all zeroes implies the oracle function is ...

- a) balanced
- b) constant
- c) periodic
- d) parity-dependent

In Deutsch–Jozsa, measuring all zeros on the counting register implies the oracle is constant (balanced functions yield a state orthogonal to $|0^n\rangle$ after the final Hadamards).

4. Grover's optimal number of iterations for one marked item in a list of size N is approximately

- a) $\sqrt{N}$
- b) $\pi\sqrt{N}/4$
- c) $N/2$
- d) $\log_2 N$

Optimal Grover iterations are the nearest integer to $(\pi/4)\sqrt{N}$

5. A depolarising channel acting on a single qubit with probability p replaces the qubit state with ...

- a) $|0\rangle$
- b) the maximally mixed state
- c) its orthogonal complement
- d) a classical bit string

Single-qubit depolarizing channel: $\mathcal{E}(\rho) = (1-p)\rho + p\,\frac{I}{2}$. With probability p, the state is replaced by $I/2$.

6. Which Qiskit transpiler optimisation level performs noise-adaptive qubit layout **and** basic gate commuting?

- a) 0
- b) 1
- c) 2
- d) 3

Qiskit optimization level 3 performs the most aggressive passes, including noise-adaptive layout and advanced commutation/scheduling; 0–2 are lighter.

7. The Quantum Fourier Transform on n qubits requires exactly ...

- a) n Hadamards and no controlled gates
- b) n controlled-phase gates and no Hadamards
- c) n Hadamards and $\frac{n(n-1)}{2}$ controlled-phase gates
- d) 2^n gates total

The standard QFT on n qubits uses one H per qubit and $\sum_{k=1}^{n-1} k = n(n-1)/2$ controlled-phase gates (ignoring final swaps if not counted).

8. In QAOA, the mixer Hamiltonian $H_B = \sum_i X_i$ is used primarily to

- a) encode the cost function
- b) initialise the ground state
- c) explore the search space
- d) measure the solution

In QAOA, the mixer $H_b = \sum_i X_i$ drives transitions among bitstrings, exploring the feasible space; the cost is encoded in H_C.

9. Over-rotating past the optimal angle in Grover's algorithm causes the success probability to

- a) stay maximal
- b) decrease then oscillate
- c) decrease monotonically to zero
- d) remain at 50 %

Grover's success probability is sinusoidal in the iteration count; overshooting the optimum causes it to drop and then oscillate.

10. The controlled-U^{2^k} gates in Quantum Phase Estimation encode the unknown phase into

- a) amplitudes of the eigenstate register
- b) relative phases of counting qubits
- c) classical ancilla bits
- d) measurement basis choice

In QPE, controlled-U^{2^k} imprint the eigenphase onto the *relative phases* of the counting register, which Hadamards+inverse QFT decode.

B1. *(Deutsch–Jozsa)* For $n = 3$, write the explicit state of the counting register **before** the final Hadamard layer when the oracle is balanced and the input is the uniform superposition produced by $H^{\otimes 3}$. Show your steps.

Start with counting register $|0\rangle^{\otimes 3}$ and ancilla $|1\rangle$. Apply $H^{\otimes 3}$ to the counting qubits and H to the ancilla to get

$$\frac{1}{\sqrt{8}} \sum_{x \in \{0,1\}^3} |x\rangle \otimes |-\rangle, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

The oracle U_f acts as $|x\rangle|y\rangle \mapsto |x\rangle|y \oplus f(x)\rangle$. With the ancilla in $|-\rangle$, this yields phase kickback:

$$U_f\left(|x\rangle|-\rangle\right) = (-1)^{f(x)}|x\rangle|-\rangle.$$

Hence, **before** the final $H^{\otimes 3}$ on the counting register, its reduced state is

$$\boxed{|\psi\rangle_{\text{count}} = \frac{1}{\sqrt{8}} \sum_{x \in \{0,1\}^3} (-1)^{f(x)}|x\rangle.}$$

For a **balanced** f, exactly four terms carry phase $+1$ and four carry -1. (Which bitstrings get the minus sign depends on the particular balanced function.)

B2. *(Grover depth)* Given $N = 16$ database items and one marked solution, a) Compute the optimal integer number of Grover iterations k_{opt}. b) Sketch how the success probability evolves for $k = 0, 1, \ldots, k_{\text{opt}} + 1$. Label the values on your sketch qualitatively (e.g., "≈1", "≈0").

a) $\sqrt{N} = 4$. The optimal iterations are $\approx (\pi/4)\sqrt{N} = \pi \approx 3.14$. Taking an integer, $\boxed{k_{\text{opt}} = 3}$ (usual choice $\lfloor \cdot \rfloor$ or nearest integer).

b) The success probability after k iterations is $P_k = \sin^2\big((2k+1)\theta\big)$ with $\sin\theta = 1/\sqrt{N} = 1/4$. Qualitatively:

- $k = 0$: small, $\approx 1/16$.
- $k = 1$: increases (order 0.4–0.5).
- $k = 2$: near maximal (≈ 0.95).
- $k = 3$: ≈ 1 (peak).
- $k = 4$: overshoots and decreases again.

  So it rises from ≈ 0 to ≈ 1 by $k = 3$, then starts oscillating down.

B3. *(Noise model)* Derive the Kraus operators for a single-qubit pure dephasing channel that leaves computational-basis populations unchanged but multiplies off-diagonal terms by e^{-t/T_2}. Clearly state the physical meaning of each Kraus operator.

A single-qubit **pure dephasing** map leaves populations unchanged and scales coherences:

$$\rho = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix} \longmapsto \mathcal{E}(\rho) = \begin{pmatrix} \rho_{00} & \lambda\, \rho_{01} \\ \lambda^* \rho_{10} & \rho_{11} \end{pmatrix}, \quad |\lambda| \leq 1.$$

A standard Kraus form (taking λ real and nonnegative for pure dephasing; write $\lambda = e^{-t/T_2}$) is

$$\boxed{E_0 = \sqrt{\tfrac{1+\lambda}{2}}\, I, \qquad E_1 = \sqrt{\tfrac{1-\lambda}{2}}\, Z,}$$

which satisfy $E_0^\dagger E_0 + E_1^\dagger E_1 = I$ and produce the above action. **Physical meaning**: with some stochasticity, the qubit experiences a random Z phase flip; averaging reduces off-diagonal terms (loss of phase coherence) while leaving ρ_{00}, ρ_{11} unchanged.
If the problem intends a *pure phase factor* $\lambda = e^{-i\phi}$ (magnitude 1), that is a **unitary** Z-rotation, representable with a single Kraus operator $U = R_z(2\phi)$; no decoherence occurs in that case.

B4. *(QAOA cost expectation)* For a ring graph of four vertices with equal weights, write the MaxCut cost Hamiltonian H_C in terms of Pauli Z operators. Using a depth-$p = 1$ QAOA state $|\psi(\beta, \gamma)\rangle$, compute $\langle Z_1 Z_2 \rangle$ after applying only the mixer layer $e^{-i\beta H_B}$ (i.e., set $\gamma = 0$). Provide the expression in terms of β.

Ring graph edges: $(1,2),(2,3),(3,4),(4,1)$. A common MaxCut cost is

$$\boxed{H_C = \tfrac{1}{2}\Big(4I - (Z_1Z_2 + Z_2Z_3 + Z_3Z_4 + Z_4Z_1)\Big).}$$

With depth $p = 1$ and $\gamma = 0$, the state is

$$|\psi(\beta,0)\rangle = e^{-i\beta H_b}|+\rangle^{\otimes 4}, \qquad H_b = \sum_{i=1}^{4} X_i.$$

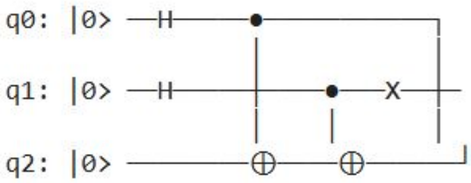
But $|+\rangle$ is an eigenstate of X with eigenvalue $+1$, so each factor $e^{-i\beta X_i}$ contributes only a **global phase** on $|+\rangle$. Therefore

$$|\psi(\beta,0)\rangle \propto |+\rangle^{\otimes 4},$$

and $\langle Z_i\rangle = 0$, $\langle Z_iZ_j\rangle = 0$. Hence

$$\boxed{\langle Z_1Z_2\rangle = 0 \quad (\text{independent of } \beta).}$$

Consider the 3-qubit circuit below (qubits indexed top→bottom). Work in the computational basis and **do not simplify with global phases**. Show the state vector **just before measurement**.



After Hadamards on q_0, q_1:

$$\frac{1}{2}\Big(|000\rangle + |010\rangle + |100\rangle + |110\rangle\Big).$$

After CNOT $q_0 \to q_2$:

$$\frac{1}{2}\Big(|000\rangle + |010\rangle + |101\rangle + |111\rangle\Big).$$

After CNOT $q_1 \to q_2$:

$$\frac{1}{2}\Big(|000\rangle + |011\rangle + |101\rangle + |110\rangle\Big).$$

After X on q_1 (flip middle bit):

$$\boxed{\frac{1}{2}\Big(|010\rangle + |001\rangle + |111\rangle + |100\rangle\Big) \;=\; \frac{1}{2}(|001\rangle + |010\rangle + |100\rangle + |111\rangle).}$$

D1. Explain why error-mitigation techniques (e.g., zero-noise extrapolation and read-out calibration) are considered stop-gaps for NISQ devices and discuss what advantages full quantum error *correction* would bring to algorithms like QPE and QAOA.

Error-mitigation techniques like zero-noise extrapolation, measurement-error calibration, and probabilistic error cancellation are *post-processing* or *circuit-level smoothing* methods that reduce bias without changing the underlying noisy device. They help in the NISQ era because they require no extra logical qubits or full syndrome extraction, but they only partially compensate errors and typically scale poorly with circuit depth and noise strength. In contrast, quantum error correction (QEC) encodes logical qubits into many physical qubits and continuously detects/corrects errors, enabling a *fault-tolerant* computation whose logical error rate can be pushed arbitrarily low by increasing code distance. Algorithms like QPE benefit dramatically: long coherent evolutions and deep controlled-unitary ladders become feasible with reliable phase readout. QAOA, when run at larger depth p, can explore richer ansätze without being throttled by decoherence and stochastic gate errors. Moreover, QEC allows standard compile-once guarantees (Clifford+T fault tolerance) and removes the need for problem-specific mitigation tuning. The price is substantial overhead in qubits, gates, and control, but the payoff is scalable, algorithm-agnostic reliability that mitigation alone cannot deliver.